Gradient Descent Algorithm

In linear regression → creating best fitting line by minimizing the error.

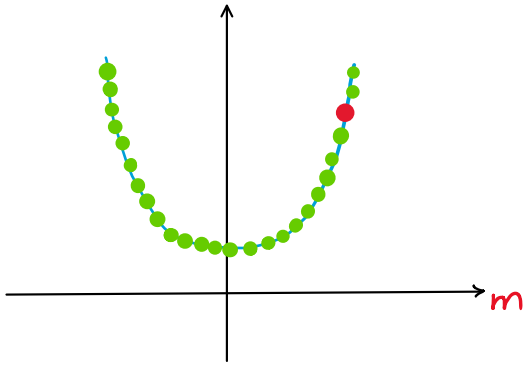
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MSE ⇝ dependent on m & c
- m → slope
- c → intercept

Finding the best value of m & c?

Assume, c=0

MSE ⇝ minimum MSE

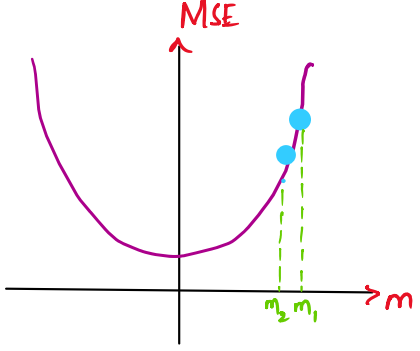


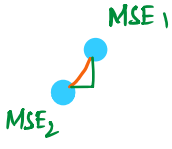
Using the above graphical representation, we want to observe change in MSE with respect to change in slope.

$\frac{\text{change in MSE}}{\text{change in slope}}$ ⇝ rate of change of MSE

differentiation

Mathematically :-





$\Rightarrow \frac{y_2-y_1}{x_2-x_1} = \frac{\Delta y}{\Delta x}$

or

$= \frac{dy}{dx} = \frac{d\,MSE}{dm}$

∴ we can say that MSE changes by changing the value of m

So,

$$\text{MSE or Loss} = \frac{\varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 + \cdots + \varepsilon_n^2}{n}$$

$$\text{Loss} = \sum_{i=1}^{n} \frac{\varepsilon_i^2}{n}$$

$$\text{Loss} = \sum_{i=1}^{n} \frac{(y_i - \hat{y}_i)^2}{n}$$

$\varepsilon = y - \hat{y}$

$\hat{y} = mx + c$

$$\text{Loss} = \sum_{i=1}^{n} \frac{(y_i - [mx_i + c])^2}{n}$$

Observing rate of change of MSE / Loss w.r.t. m

$$\frac{d\,\text{Loss}}{dm} = \frac{d}{dm}\left[\sum_{i=1}^{n} \frac{(y_i - [mx_i + c])^2}{n}\right]$$

$$\frac{dL}{dm} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{d}{dm}(y_i - [mx_i + c])^2\right]$$

$\frac{d}{dx}x^2 = 2x$

$\frac{d}{dx}y^2 = 2y \cdot \frac{dy}{dx}$

$$= \frac{2}{n}\sum_{i=1}^{n}\left[y_i - (mx_i + c)\right] \cdot \frac{d}{dm}(y_i - [mx_i + c])$$

$$= \frac{2}{n}\sum_{i=1}^{n}\left[y_i - (mx_i + c)\right] \cdot \left\{\frac{dy_i}{dm} - \frac{d\,mx_i}{dm} - \frac{dc}{dm}\right\}$$

$$= \frac{2}{n}\sum_{i=1}^{n}\left[y_i - (mx_i + c)\right] \cdot [0 - x_i - 0]$$

$$\frac{dL}{\cdot} = -\frac{2}{n}\sum_{i=1}^{n}\left[y_i - (mx_i + c)\right] \cdot x_i \quad \text{——} \quad (1)$$

$$\frac{dL}{dm} = -\frac{2}{n}\sum_{i=1}^{n}\left[y_i-(mx_i+c)\right]\cdot x_i \quad \text{——} \quad (1)$$

Similarly, by keeping $m=0$ & taking c

$$\frac{dL}{dc} = \frac{2}{n}\sum_{i=1}^{n}\left[y_i-(mx_i+c)\right]\cdot\frac{d}{dc}\left(y_i-[mx_i+c]\right)$$

$$= \frac{2}{n}\sum_{i=1}^{n}\left[y_i-(mx_i+c)\right]\cdot\frac{d}{dc}y_i-\frac{d}{dc}mx_i-\frac{d}{dc}c$$

$$= \frac{2}{n}\sum_{i=1}^{n}\left[y_i-(mx_i+c)\right]\cdot[0-0-1]$$

$$\frac{dL}{dc} = -\frac{2}{n}\sum_{i=1}^{n}\left[y_i-(mx_i+c)\right]\cdot 1 \quad \text{——} \quad (2)$$

Using eq. (1) & (2), we can calculate the change in loss/MSE



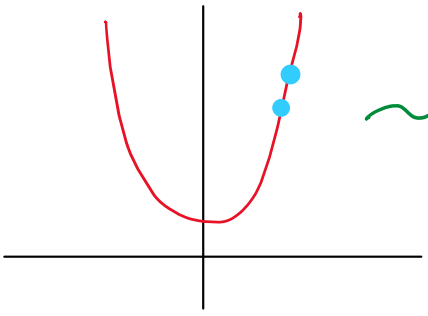
∴ for updating the value of m

$$m_{new} = m_{old} - \frac{dL}{dm}$$

$$c_{new} = c_{old} - \frac{dL}{dc}$$

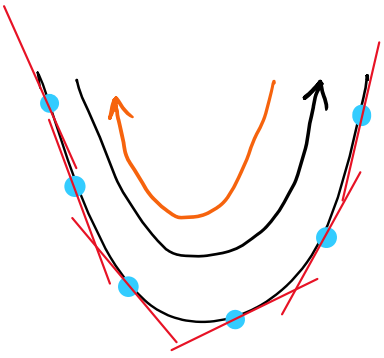
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we are subtracting it because we want to observe minimum loss/MSE



⇝ moving in upward direction will increase the error

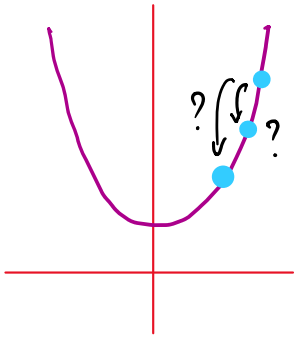
So, we will move in the opp. direction



⇝ moving in ↗ direction, will give +ve loss

moving in ↖ direction, will give -ve loss

* Learning Rate :-



⇝ what steps does the loss takes in order to reach minimum error

This is decided by step size

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learning rate

The learning rate is the step size taken in order to reach the minimum error. It is denoted by 'η'

$$\frac{dL}{dm} = 6 \quad \& \quad \eta = 0.1$$

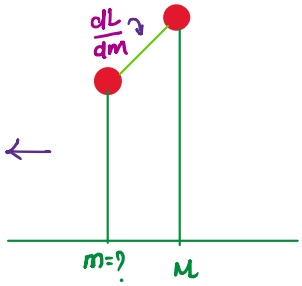
$$\frac{dL}{dm} * \eta = 6 \times 0.1 = 0.6$$

∴ the new updated m & c will be →

∴ Final eq. for updating m & c will be →

$$m_{new} = m_{old} - \eta \frac{dL}{dm}$$

$$c_{new} = c_{old} - \eta \frac{dL}{dc}$$



So, this is how the value of m & c is calculated / decided